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Cognitive process grouping on students' mathematical creative problem solving: Evidence from clustering analysis, lag sequence analysis, and hidden Markov model

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ABSTRACT

Creative problem solving (CPS) is vital for developing students' critical thinking and innovation skills. However, existing research often overlooks the heterogeneity in students' CPS behaviors, especially in mathematics, making it challenging to design effective, personalized instruction. This study addresses this gap by exploring the heterogeneity of students' behaviors in mathematical creative problem solving (MCPS) among junior school students. To achieve this, behavioral data were collected from 778 middle school students (59.6% male) during computer-based MCPS tasks, capturing rich, time-dependent problem-solving actions. The research employed a K-means clustering approach to identify distinct student profiles based on ability levels and cognitive process metrics. To further understand the behavioral patterns within these profiles, lag sequential analysis and hidden markov models were used to examine the sequences and transitions of problem-solving actions over time. The analysis revealed significant heterogeneity in students' behavioral patterns across the six profiles, providing nuanced insights into their problem-solving strategies and approaches. This study's primary contribution lies in demonstrating how combining clustering with sequence analysis of behavioral data can uncover nuanced student profiles, informing personalized learning approaches and advancing research methodologies in educational data mining. These findings also provide meaningful insights for future research aimed at developing targeted interventions and supporting the continuous improvement of CPS instructional strategies.

1. Introduction

In an increasingly innovation-driven economy, the rapid evolution of knowledge and technology necessitates that citizens become adept at navigating complex and novel situations. Accordingly, educators and researchers globally increasingly recognize the critical importance of fostering students' creative problem solving (CPS) skills for future readiness (Suherman & Vidákovich, 2022; Thornhill-Miller et al., 2023). As an essential 21st-century competency, CPS integrates creativity, critical thinking, divergent and convergent thinking, and metacognitive regulation to tackle ill-defined problems (Treffinger et al., 2023; Urban et al., 2024; Urban & Urban, 2023). CPS is often defined as an iterative approach employing unconventional thinking to develop unique and valuable solutions for

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3. Methods

3.1. Participants and procedure

Participants in this study comprised a total of 808 Chinese students from grades 7 to 9, recruited via convenience sampling from a public junior high school in Shanxi Province, a region with a moderate level of economic and educational development. All participants were tested as intact classes, with no prior screening based on academic performance or other criteria, to ensure the sample reflected the natural distribution of students at these grade levels. Before this study, students had limited prior experience with formal computer-based mathematics assessments but possessed basic computer skills acquired through regular school IT courses. To minimize the influence of interface unfamiliarity, a training session on the test system was conducted prior to the formal assessment. The initial data screening involved excluding responses from students who answered fewer than three questions, resulting in a final sample of 778 students (314 females, 464 males). This step ensured data quality by removing incomplete or potentially invalid responses.

Prior to data collection, all participants were informed about the purpose and procedures of the study. To ensure ethical compliance, the proctor read aloud the informed consent statement displayed on the test system's interface before the commencement of the assessment. Participants indicated their voluntary participation by explicitly agreeing to proceed with the test, thereby providing informed consent in an online environment. During the administration of the online test, the proctor and administrators closely monitored the process to provide necessary technical assistance. They addressed any system usage issues that arose, such as login difficulties, navigation problems, or technical glitches, and recorded these incidents to analyze potential impacts on participants' performance.

3.2. Assessment instrument

The assessment tool used in this study is grounded in the Mathematical Creative Problem Solving (MCPS) cognitive model proposed by Tan (2024) and Tan et al. (2026). This model synthesizes established theories of general creative problem-solving (e.g., Mumford et al., 1991; Treffinger et al., 2023) with domain-specific demands of mathematics (e.g., Kim et al., 2004). It conceptualizes MCPS as a dynamic process comprising three primary phases: Problem Finding, Problem Solving, and Monitoring & Reflection, which are further delineated into seven secondary dimensions. This comprehensive framework, detailed in Table 1, served as the blueprint for developing the computer-based interactive tasks and for analyzing the fine-grained cognitive processes in this study.

Based on a cognitive model of MCPS (see Table 1), Tan (2024) developed two computer-based assessment tasks that are psychometrically equivalent: “Buying Movie Tickets during the May Day Holiday” and “Camping Tent Rental”. These tasks were designed to evaluate MCPS skills within a realistic and engaging context, allowing for the measurement of various cognitive processes. A detailed description of both assessment tasks (“Buying Movie Tickets during the May Day Holiday” and “Camping Tent Rental”), including their structure, item formats, and illustrations of the system interface, is provided in Appendix A.

Furthermore, Appendix B presents illustrative sample student responses. Each of the two tasks comprises six sequentially structured items that are designed to progressively build upon prior responses, thereby capturing the dynamic nature of problem-solving processes. The item formats include short answer questions, multiple-choice questions, dropdown menu selections, and scroll-based interactions, providing a comprehensive assessment of participants’ reasoning, decision-making, and metacognitive regulation across different problem-solving stages. The tasks are timed, with an approximate completion duration of 45 to 60 min for both activities, ensuring sufficient depth of engagement while maintaining feasibility for participants.

The overall internal consistency, measured by Cronbach's α across all items, was 0.78. This value indicates acceptable to good reliability for the purposes of this study (DeVellis, 2017). While it falls slightly below the more stringent threshold of 0.80 often recommended for confirmatory research (Nunnally & Bernstein, 1994), it is well above the 0.70 benchmark commonly accepted for exploratory and basic research, suggesting the items collectively measure the target construct with adequate consistency. Regarding the two items with weaker factor loadings (as reported in the CFA below), theoretical considerations were prioritized over statistical optimization. These items were retained because they capture essential aspects of MCPS—such as metacognitive monitoring and flexible information selection in ill-defined contexts—that are not fully represented by other items. Removing them would compromise construct coverage by narrowing the breadth of the measurement. The current factor structure, despite modest loadings for a few items, provides a more complete operationalization of the MCPS framework (Tan, 2024; Tan et al., 2026) than a statistically optimized but theoretically truncated solution.

Furthermore, a confirmatory factor analysis (CFA) was performed to examine the construct validity and the factorial structure of the assessment. The CFA results demonstrated a good model fit, with $\chi^2/df = 2.21$, $RMSEA = 0.06$, $CFI = 0.90$, and $SRMR = 0.08$. These indices indicate that the data fit the hypothesized model well, supporting the alignment of the assessment with the underlying theoretical framework. Additionally, the factor loadings ranged from 0.38 to 0.88. According to conventional benchmarks (e.g., Comrey & Lee, 1992; Tabachnick & Fidell, 2019), two items had factor loadings of 0.38, indicating weak associations. The remaining items had loadings ranging from 0.45 to 0.88, which reflect moderate to strong associations with the latent construct of CPS ability. This pattern further corroborates the construct validity of the tool, confirming that the items collectively capture different facets of MCPS skills, despite one item exhibiting a weaker relationship with the overall construct.

3.3. Data collection and datasets

This study systematically collected fine-grained process data to capture participants' complete interactions during the computer-

based assessment. All actions were recorded with timestamps across nine defined categories: subject ID, item code, operation type (e.g., click, input), operation area (e.g., response area, interaction area), operation object (e.g., buttons, information elements), operation result, response content, operation name, and timestamp. This created a chronological log for each participant, which was then organized into a structured time-action sequence dataset. To transform the raw interaction logs into quantifiable data for analysis, a systematic coding framework was applied.

Table 2 illustrates this scheme using Question 1 of the "Buying Movie Tickets during the May Day Holiday" task as an example. Each recorded action was categorized into key dimensions: Item code, Operation type, Operation area, Operation subject, Operation result, Response content, and a standardized Operation name (e.g., info1, submit1). This process converted continuous logs into discrete, labeled event sequences, enabling the reconstruction of individual problem-solving trajectories—including how students navigated information, used tools, and revised answers. The resulting formatted time-action sequence dataset serves as the essential input for the advanced analytical techniques central to this study, namely K-means time-series clustering and LSA, allowing for the investigation of behavioral patterns and strategies at the process level.

In addition to behavioral data, response content was classified into multiple-choice (including fill-in-the-blank) and short-answer questions, with responses recorded as response codes or written responses accordingly. For multiple-choice questions, response codes were documented, while written responses were recorded for short-answer questions. These response data served as the basis for scoring participants' performance across seven secondary dimensions by Tan (2024), in accordance with established scoring criteria. Each participant's responses were evaluated to generate seven-dimension scores, as well as an overall total score, which provided a comprehensive measure of MCPS abilities. To ensure comparability across different dimensions and facilitate interpretation, raw scores were converted into percentage-based score rates by multiplying the raw scores by 100. This normalization process resulted in standardized scores where higher percentage values indicated better performance in the respective MCPS dimensions. The outcome data, including these standardized scores, were organized into a dedicated outcome dataset, enabling subsequent statistical analysis and interpretation of participants' MCPS ability. This multi-layered data collection approach allowed for an in-depth examination of both behavioral processes and performance outcomes, providing a robust foundation for understanding how participants engage with complex problem-solving scenarios within the system.

To establish scoring consistency and reliability, three trained raters randomly selected ten student responses for joint scoring sessions. During these sessions, raters collaboratively scored the responses, discussed discrepancies, and refined scoring rubrics to ensure a shared understanding of scoring standards. This joint calibration process aimed to minimize rater biases and enhance inter-rater reliability. Following this calibration, the raters independently scored a larger set of responses, documenting their scoring procedures, criteria, and decisions throughout the process. They carefully addressed any items deemed unscorable—such as responses that were irrelevant, illegible, or lacked sufficient detail—and made necessary adjustments to the scoring criteria as needed.

3.4. Data mining procedure

Data mining was performed in a Python 3 environment using the *scikit-learn*, *talearn*, *statsmodels*, and *hmmlearn* libraries. The analysis began with descriptive statistics of the overall sample, including tests for gender and grade-level differences in MCPS performance.

A two-stage hierarchical clustering strategy was adopted rather than a single-stage solution (e.g., directly clustering all students into several groups) for three interrelated reasons. First, ability-level clustering was conducted first because overall mathematical competence is a primary determinant of MCPS performance; partitioning students by ability establishes macro-level strata that reflect domain-general constraints on problem-solving. Second, behavioral sequence clustering was performed within each ability stratum to uncover process heterogeneity that would otherwise be masked by ability differences. Students with similar competence may employ qualitatively different strategies (e.g., efficient vs. reflective), and holding ability constant allows these fine-grained patterns to emerge. Third, this hierarchical approach prevents ability variance from dominating process-pattern detection. If a single-stage clustering were applied directly, the resulting groups would likely be driven primarily by ability differences, obscuring within-ability process distinctions. By separating ability and process variance, the two-stage design ensures that the final subclusters reflect genuine behavioral heterogeneity rather than artifacts of ability-driven partitioning.

Table 2
Example of process data collection and coding rules.

Item code	Operation type	Operation area	Operation subject	Operation result	Response content	Operation name
A1	Click	Interaction	Chat history	Show chat history		chat
A1	Click	Interaction	Information center	Show information center		info
A1	Click	Interaction	Information	Show information		Info1∼7
A1	Click	Note	Add button	Add note box		addnote1
A1	Click	Note	Delete button	Delete note box		deletenote1
A1	Click	System	Submit button	Submit answer and notes	XXXX	submit1
A1	Click	System	Next button	End this item		shift12
A1	Click	System	Item 3	Enter item 3		shift13
A1	Click	System	Item 4	Enter item 4		shift14
A1	Click	System	Item 5	Enter item 5		shift15
A1	Click	System	Item 6	Enter item 6		shift16

Building on this rationale, the approach then involved a two-stage K-means clustering process aimed at identifying both performance-based and behavior-based student groups, complemented by LSA to examine behavioral patterns in detail. The first stage focused on clustering students based on their performance scores across seven CPS dimensions in mathematics. To prepare the data, raw scores across the seven CPS dimensions in mathematics were extracted for each student and then standardized using z-score transformation, converting each variable to have a mean of 0 and a standard deviation of 1. This process ensured that each variable contributed equally to the clustering process and prevented biases due to differing scales. Data quality checks were performed to identify and address any missing or outlier scores, ensuring a clean and reliable dataset. Next, the optimal number of clusters (K) was determined by calculating the Silhouette Score for a range of K values, typically from 2 to 10. The Silhouette Score measures how well each student fits within their own cluster relative to other clusters, with higher scores indicating better-defined clusters (Kodinariya & Makwana, 2013). The value of K corresponding to the maximum average Silhouette Score was selected as the optimal cluster number. To validate this choice, additional metrics such as the Davies-Bouldin Index and the Calinski-Harabasz Index were computed, with the former favoring lower values and the latter favoring higher values for better cluster separation and cohesion (Ros et al., 2023; Wang & Xu, 2019).

Following the selection of K, the K-means algorithm was executed with multiple initializations to ensure robustness and to avoid local minima. In each iteration, students were assigned to the nearest centroid based on Euclidean distance, and the centroids were recalculated as the mean of assigned students. This process repeated until the centroids stabilized, indicating convergence. The clustering solution with the best validation metrics was retained for further analysis. To assess the quality and stability of the clusters, multiple runs with different random seed initializations were performed, and the consistency of cluster assignments was checked. Lastly, the interpretation of the performance-based clusters involved examining the mean scores and distributional characteristics across the seven dimensions (Ikotun et al., 2023).

The second stage involved analyzing behavioral patterns through K-means time-series clustering of student action sequences. The raw action data were encoded into sequences reflecting the order and type of actions performed during the tasks. These sequences were formatted into appropriate time-series data structures, with categorical actions encoded numerically or categorically. The data were then standardized to ensure comparability. Using the talearn library's time-series K-means algorithm, students were grouped based on their behavioral trajectories. Similar to the first stage, multiple K values were tested, and the optimal number of behavioral clusters was selected based on validation metrics such as the Silhouette Score (Kodinariya & Makwana, 2013). Multiple initializations were run to ensure cluster stability and reliability. The validation of behavioral clusters used the same set of indices—Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index—to assess the quality of the clustering solutions (Ros et al., 2023; Wang & Xu, 2019). Consistency across multiple runs was checked to confirm stability.

The third stage is to use LSA to examine behavioral patterns (Bakeman & PhD, 1997). For above each cluster, a transition frequency table was generated, capturing how often specific actions followed one another within student sequences. Residual analysis was then performed on these transition frequencies to identify significant behavioral shifts. Expected transition frequencies were calculated assuming independence between actions, and residuals (Z-scores) were computed to measure deviations from these expectations. Transitions with Z-scores exceeding 1.96 were considered statistically significant, indicating meaningful behavioral pathways. Finally, significant behavioral transitions were visualized through directed graphs or pattern diagrams, illustrating the common sequences and trajectories within each cluster. These visualizations provided insights into typical response behaviors, decision points, and problem-solving strategies characteristic of different student groups, enriching the understanding of the underlying behavioral processes.

The fourth stage applied HMMs to quantify latent stage dynamics (Fine et al., 1998). HMMs were fitted separately to each of the six behavioral subclusters, using the same action sequences as observed variables. The number of hidden states for each subcluster was selected by minimizing the Bayesian information criterion (BIC) over 2–15 states. Parameter estimation used the Baum-Welch algorithm. For each subcluster, we extracted state stability and mean duration, the transition probability matrix, emission probabilities, and the steady state distribution. To compare transition structures across clusters, we performed state matching based on emission probabilities (Hungarian algorithm) for each pair of subclusters within the three main clusters. The matched state pairs and differences in stability, mean duration, and steady-state proportion are reported.

Table 3
Descriptive statistics and inter-dimensional differences in mathematical creative problem-solving (MCPS) performance.

Dimensions	Mean	SD	1	2	3	4	5	6	7
1 Problem Identification	62.66	39.02	-	25.20***	20.01***	34.74***	31.35***	37.14***	39.32***
2 Information Selection	24.52	14.88	0.09*	-	−9.66***	14.45***	5.95***	23.60***	24.01***
3 Variable Identification	32.16	20.31	0.10*	0.21***	-	20.73***	12.72***	27.51***	29.42***
4 Solution Generation	10.82	22.63	0.16***	0.05	0.11**	-	−10.59***	0.90	3.55***
5 Solution Evaluation	19.18	20.38	0.27***	0.02	0.01	0.46***	-	11.79***	15.88***
6 Monitoring &Adjustment	10.09	9.18	0.06	0.05	0.02	0.19***	0.09***	-	4.04***
7 Expansion &Reflection	8.02	13.05	0.18***	0.08*	0.12**	0.30***	0.36***	0.19***	-

Note. The lower-left section showed the correlation coefficients between the dimensions, while the upper-right section displayed the t-values from the paired sample t-tests between the dimensions.
*$p < .05$. **$p < .01$.***$p < .001$.

4. Results

4.1. Descriptive statistics analysis

The average scores of all students across various dimensions of creative problem solving ability ranged from 8.02 to 62.66. According to Table 3, results from a Multivariate Analysis of Variance (MANOVA) indicated significant differences in performance across the seven dimensions ($F = 865.54$, $p < .001$, *Wilks' Lambda* $= 0.11$). Post hoc analysis revealed that, with the exception of Solution Generation and Monitoring & Adjustment, all other dimensions showed significant differences between pairs (see Table 3). Specifically, students' performance levels, ranked from highest to lowest, were as follows: Problem Identification, Variable Identification, Information Selection, Solution Evaluation, Solution Generation and Monitoring & Adjustment, and Expansion & Reflection.

As shown in Table 4, the results of the independent samples *t*-test showed that there were no significant differences between boys and girls in six of the seven dimensions of creative mathematical problem solving, but girls demonstrated significantly higher levels of Expansion and Reflection than boys ($t = 3.47$, $p = .001$), with a small effect size ($d = 0.25$).

As shown in Table 5, the results of the one-way analysis of variance (ANOVA) showed that there were no statistically significant differences across grades 7 to 9 in six of the seven secondary dimensions of mathematical creative problem solving (F values ranging from 0.86 to 1.69, all $p > .05$). The exception was found for Monitoring & Adjustment, where a significant main effect of grade was observed ($F = 5.14$, $p = .006$). Post-hoc comparisons indicated that 9th graders scored significantly higher on this dimension compared to 7th and 8th graders. However, the effect size for this difference was small ($\eta^2 = 0.01$).

4.2. K-means clustering analysis

This first-stage analysis aimed to identify distinct student profiles based on their overall performance across the seven MCPS dimensions. K-means clustering was applied to the standardized multi-dimensional ability scores, with the optimal number of clusters determined via silhouette analysis. After data preparation, silhouette analysis was used to determine the optimal number of clusters (K). As shown in Table 6, the results indicated that K = 3 produced the best clustering outcome, with a silhouette coefficient of 0.34, which suggested that three clusters provided a reasonable balance between intra-cluster cohesion and inter-cluster separation.

Additionally, one-way ANOVA was conducted to examine the differences in CPS performance across the three clusters. The results showed significant performance differences across clusters for all seven dimensions (see Table 7). Medium effects were found for Problem Identification and Monitoring & Adjustment ($0.14 > \eta^2 > 0.06$), while the other five dimensions had large effects ($\eta^2 > 0.14$). Post hoc tests using Dunnett's T3 method confirmed significant differences across clusters for all dimensions.

As shown in Table 7, *Creative Problem Solvers* (Cluster 1, 18.64% of students) excelled across all seven dimensions, demonstrating strong problem finding and solving skills and high metacognitive abilities. *Underperformers* (Cluster 2, 70.44% of students), the largest group, had the lowest abilities in all dimensions. They struggled with problem finding and solving and lacked the ability to monitor and adjust. *Problem Finders* (Cluster 3, 10.92% of students) excelled in the first three dimensions but did average in the last four dimensions, which indicated that these students were strong in understanding problems and analyzing variables but struggled with completing the full problem-solving process.

To validate the clustering results, several metrics were calculated. The Silhouette Score of 0.38 indicated a moderate level of similarity within clusters, though some overlap exists between adjacent clusters. The Davies-Bouldin Index of 1.39 suggested a moderate level of separation between clusters, with some degree of overlap. Finally, the Calinski-Harabasz Index of 210 demonstrated that the between-cluster variance is greater than the within-cluster variance, further supporting the validity of the clustering model. Overall, the K-means clustering results were of good quality, with clear separation between the groups and moderate within-cluster similarity.

Table 4
Gender differences in mathematical creative problem-solving (MCPS) ability across secondary dimensions.

Dimensions	Gender	N	Mean	SD	*t*	*p*	*d*
1 Problem Identification	Female	300	65.67	37.32	1.76	0.079	0.13
	Male	464	60.67	39.97			
2 Information Selection	Female	300	23.27	12.83	−1.21	0.225	0.09
	Male	464	24.48	14.54			
3 Variable Identification	Female	300	30.47	17.95	−1.47	0.142	0.11
	Male	464	32.63	20.99			
4 Solution Generation	Female	300	11.33	22.18	0.87	0.387	0.06
	Male	464	9.91	22.07			
5 Solution Evaluation	Female	300	19.93	19.87	0.91	0.363	0.07
	Male	464	18.58	20.46			
6 Monitoring & Adjustment	Female	300	10.61	9.78	1.42	0.157	0.11
	Male	464	9.62	8.78			
7 Expansion & Reflection	Female	300	9.80	14.02	3.47**	0.001	0.25
	Male	464	6.59	11.33			

Note. *$p < .05$. **$p < .01$. ***$p < .001$.

Table 5
Grade differences in mathematical creative problem-solving (MCPS) ability across secondary dimensions.

Dimensions	Grade	*N*	M	SD	*F*	*p*	η^2
1 Problem Identification	7	418	61.24	39.85	1.32	0.268	0.00
	8	216	66.20	36.30			
	9	130	60.77	40.34			
2 Information Selection	7	418	24.78	14.40	0.99	0.373	0.00
	8	216	23.15	11.94			
	9	130	23.85	16.72			
3 Variable Identification	7	418	30.91	18.57	0.86	0.424	0.00
	8	216	32.96	21.45			
	9	130	32.46	21.17			
4 Solution Generation	7	418	9.09	20.95	1.69	0.186	0.00
	8	216	11.85	22.50			
	9	130	12.31	24.70			
5 Solution Evaluation	7	418	19.86	18.97	1.53	0.218	0.00
	8	216	19.07	23.45			
	9	130	16.31	18.26			
6 Monitoring & Adjustment	7	418	9.24	7.63	5.14**	0.006	0.01
	8	216	10.02	10.71			
	9	130	12.18	10.54			
7 Expansion & Reflection	7	418	7.89	12.07	1.35	0.259	0.00
	8	216	6.94	13.01			
	9	130	9.23	13.21			

Note. *$p < .05$. **$p < .01$. ***$p < .001$.

Table 6
Silhouette coefficients for K-means clustering analysis.

K	2	3	4	5	6	7
Silhouette coefficient	0.33	0.34	0.22	0.26	0.26	0.29

Table 7
The results of one-way ANOVA.

Dimensions	Creative Problem Solvers (Cluster 1, *N* = 145)		Underperformer (Cluster 2, N = 548)		Problem Finders (Cluster 3, N = 85)		*F*	η^2	Dunnett's T3
	M	SD	M	SD	M	SD			
1 Problem Identification	81.03	29.50	56.48	40.33	71.18	33.07	26.63***	0.06	Cluster 1 = Cluster 3> Cluster 2
2 Information Selection	23.59	9.91	21.90	8.25	43.06	32.29	92.32***	0.19	Cluster 3> Cluster 1 = Cluster 2
3 Variable Identification	32.55	19.57	25.33	9.09	75.53	20.33	531.99***	0.58	Cluster 3> Cluster 1> Cluster 2
4 Solution Generation	46.90	26.60	1.72	8.25	8.00	16.96	557.39***	0.59	Cluster 1> Cluster 3> Cluster 2
5 Solution Evaluation	45.52	24.44	12.81	13.31	15.29	13.68	242.16***	0.38	Cluster 1> Cluster 2 = Cluster 3
6 Monitoring &Adjustment	15.09	12.43	8.90	7.64	9.24	9.01	28.317***	0.07	Cluster 1> Cluster 2 = Cluster 3
7 Expansion &Reflection	20.55	17.79	4.27	8.72	10.82	12.93	119.25***	0.24	Cluster 1> Cluster 3> Cluster 2

Note. *$p < 0.05$, **$p < 0.01$, and ***$p < 0.001$.

4.3. Time-series K-means clustering analysis

The second-stage analysis sought to uncover behavioral heterogeneity within each primary ability profile identified above. For each of the three main clusters, a separate time-series K-means clustering was performed on the fine-grained time-action sequence data to group students based on the similarity of their problem-solving processes and temporal action patterns. Silhouette analysis was employed to determine the optimal number of clusters within each of the three primary ability-based profiles.

The results of the time-series K-means clustering revealed two distinct behavioral sub-clusters within each main cluster (see Figs. 1–3). The silhouette coefficients for these sub-cluster solutions were 0.34 (*Creative Problem Solvers*, Cluster 1), 0.42 (*Underperformers*, Cluster 2), and 0.36 (*Problem Finders,* Cluster 3), indicating a moderate level of internal cohesion. The characteristic action sequences of each sub-cluster are visualized in Figs. 4–6, where the x-axis shows the action sequence index, the y-axis represents the coded behaviors, gray lines plot the raw behavioral paths of individual students, and the red line denotes the cluster centroid (the prototypical sequence).

Specifically, within *Creative Problem Solvers* (Cluster 1), Cluster 1.1 exhibited a short response duration with sparse behavioral transitions (slower pace of change), whereas Cluster 1.2 demonstrated a longer response time despite a similarly sparse sequence pattern (see Fig. 4). Within *Underperformers* (Cluster 2), Cluster 2.1 was marked by a long response time and dense action transitions

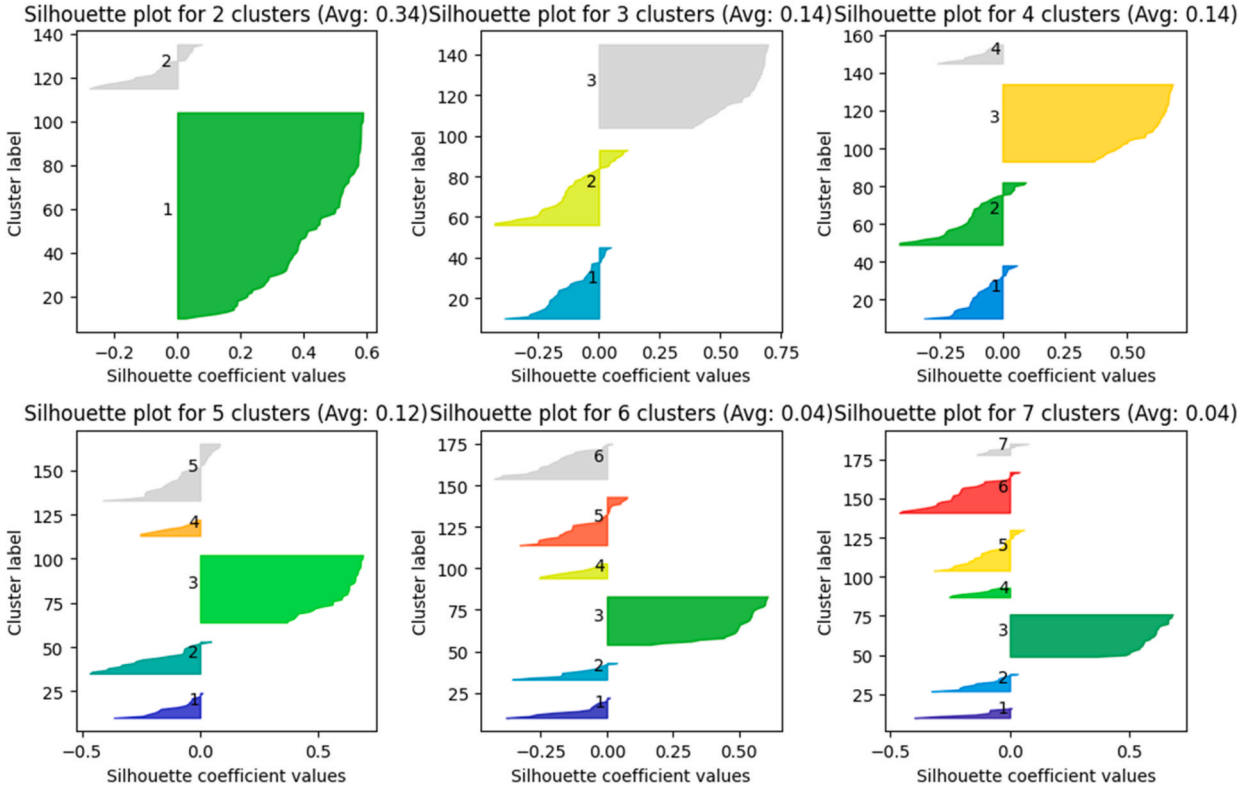

Fig. 1. Silhouette plots for cluster 1.

(rapid, frequent changes), in contrast to Cluster 2.2, which had a shorter duration but equally dense lines (see Fig. 5). Within *Problem Finders* (Cluster 3), Cluster 3.1 showed a short response time with dense transitions (fast-paced changes), while Cluster 3.2 was characterized by a longer response time and sparser lines, suggesting a more deliberate, slower-paced approach (see Fig. 6).

The validity of this sub-clustering structure was further supported by complementary metrics. Davies-Bouldin Indexes of 2.05, 1.68, and 1.76 for three clusters suggested moderate separation between clusters. Calinski-Harabasz Indexes of 20.36, 111.19, and 16.67 for three clusters showed greater between-cluster variance, confirming the model's validity. Collectively, these indices confirm a meaningful and valid partition of behavioral sequences within each primary profile, indicating good overall clustering quality.

4.4. Lag sequence analysis

To analyze the action sequence patterns of the six subclusters, a lag sequential analysis of students' action sequences identified typical patterns, with residual and behavior pattern diagrams detailed in Appendix C. The behavioral transition residual diagrams were obtained from the behavior sequence dataset, with rows and columns corresponding to different behaviors, where the rows represented the preceding behavior and the columns represented the subsequent behavior (in the order they appear in the dataset). The darker the color, the higher the likelihood of the subsequent behavior occurring after the preceding behavior. The behavioral pattern diagrams illustrated relationships between actions through nodes, which represent behaviors (with larger diameters indicating higher likelihoods of occurrence), edges, which signify transitions between behaviors (with thickness reflecting the strength of association), and arrows, which depict the sequence of actions, while applying an edge weight threshold highlights significant patterns by filtering out weaker transitions to reveal key behavioral characteristics and group differences.

The features of the behavioral pattern diagrams for six subclusters were summarized in Table 8. Cluster 1.1 and Cluster 1.2 had 58 and 56 nodes with 156 and 153 edges, respectively, and similar densities (average degree: 5.38 vs. 5.46). Cluster 1.1 showed higher outputs (7 for *shift21*) and inputs (8 for *shift21*) compared to Cluster 1.2 (6 for *shift12/22/23/34/44*; 7 for *shift12/34*) with greater activity in getting information, shifting items, and submitting answers.

Cluster 2.1, the largest group (57.66%), showed the most complex network with 58 nodes, 189 edges, the highest average degree (6.52), the highest maximum output (9 for *shift56*) and input (8 for *shift11/65*), indicating extensive interactions and intense, task-specific behaviors. Cluster 2.2 (14.25%) was similarly complex, with 58 nodes, 180 edges, an average degree of 6.21, and notable maximum output (8 for *shift34*) and input (7 for *shift44*).

Cluster 3.1 (3.60%) was the simplest, with 48 nodes, 99 edges, and the lowest average degree (4.13), showing minimal interactions and engagement. Cluster 3.2 (6.52%) featured a moderately simple network with 52 nodes and 125 edges, showing limited

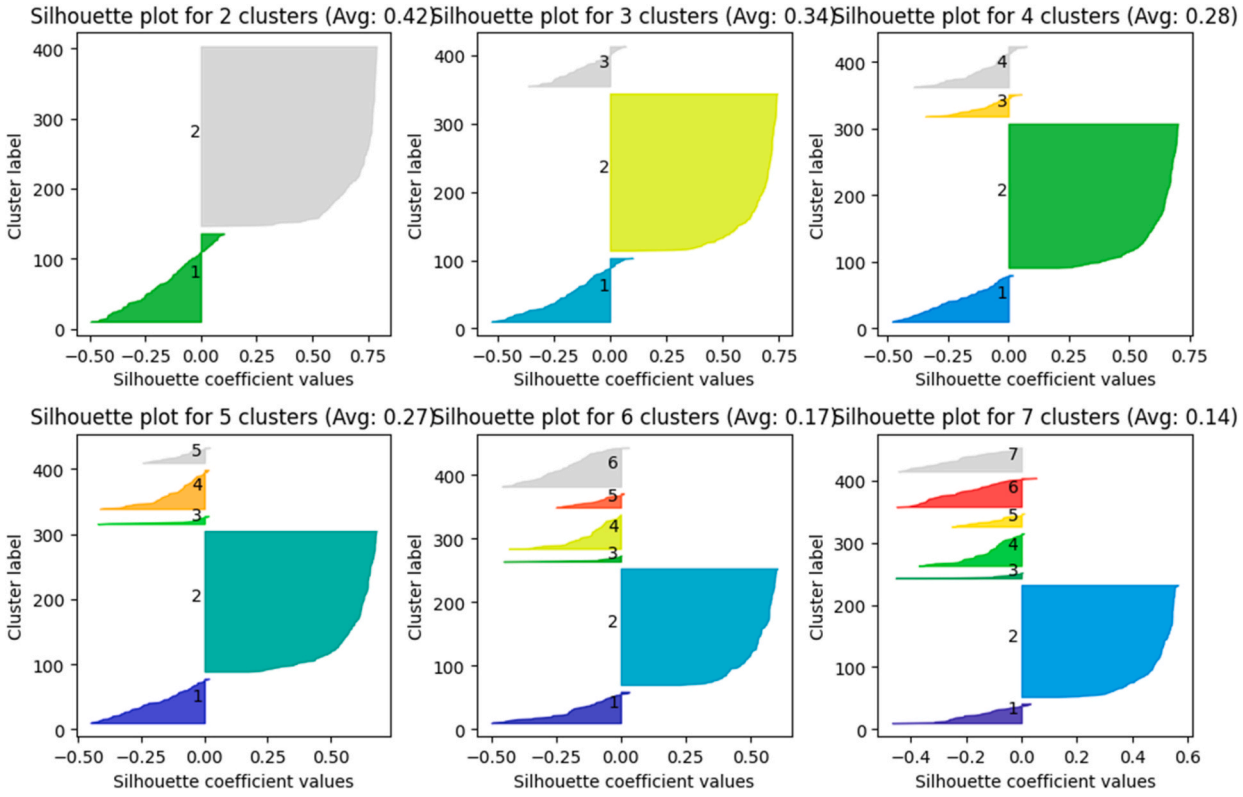

Fig. 2. Silhouette plots for cluster 2.

interactions with an average degree of 4.81, a maximum output of 6 for *shift44*, a maximum input of 7 for *shift12*, and a slightly more active but still less dynamic pattern compared to Clusters 2.1 and 2.2.

When the edge weight threshold for the pattern diagrams was increased to 15, distinct action sequence characteristics of the two student groups became evident, as summarized in Table 9. Clusters 1.1, 1.2, and 2.1 exhibited consistent workflows, such as sequential task submissions (e.g., *submit1* → *shift12* → *submit2* → *shift23*) and logical task progression toward task completion (e.g., *submit4* → *shift45*, *submit6* → *taskend*), which suggested a structured and goal-oriented task approach. Clusters 1.2 and 2.1 shared additional complexities, like incorporating chat → info interactions, emphasizing communication and information exchange in these clusters.

As for the distinct patterns, Cluster 1.1 featured unique bidirectional shifts (e.g., *shift14* → *shift41*), indicating a misunderstanding of the problem context. Cluster 1.2 showed varied interactions (e.g., *chat* → *info*) and complex shifts (e.g., *shift65* → *shift55*), reflecting multitasking. Cluster 2.1 highlighted strategic organization through note-taking (e.g., *addnote* → *deletenote*) and extended info sequences (e.g., *info2* → *info5*). Clusters 3.1 and 3.2 exhibited simpler patterns, with Cluster 3.2 focusing on shifts (e.g., *submit5* → *shift56*). Therefore, Clusters 1.2 and 2.1 showed complex, interactive behaviors with higher task engagement, while Clusters 3.1 and 3.2 exhibited simpler patterns, indicating reduced interaction.

4.5. Hidden markov models

Based on the HMM analyses, the best-fitting number of hidden states varied across subclusters (Table 10). The full model selection results are provided in Appendix D. Across clusters, state stability ranged from 0.27 to 0.98, and mean durations from 1.37 to 64.23 steps. Complete emission probabilities, stability, duration, and steady-state proportions per state are given in Appendix E. Transition graphs and matrices are in Appendix F.

To compare subclusters within each pair (Clusters 1.1 vs. 1.2, Cluster s2.1 vs. 2.2, Clusters 3.1 vs. 3.2), we applied a state-matching procedure based on emission probabilities (Hungarian algorithm). The matched state pairs and differences in stability, mean duration, and steady-state proportion are summarized in Tables 11–13. For the Cluster 1 pair, three matched pairs showed high similarity (cosine distance <0.21). Differences consistently favored Cluster 1.1, with one state in Cluster 1.2 remaining unmatched. For the Cluster 2 pair, the closest match had a distance of 0.12; four states in Cluster 2.1 had no counterpart in Cluster 2.2. For the Cluster 3 pair, the largest differences were observed for one matched pair ($\Delta\text{Steady} = +0.437$), and one state in Cluster 3.2 was unmatched. Overall, the state-matching analysis revealed systematic differences in dynamic structure between the two subclusters within each of the three main clusters.

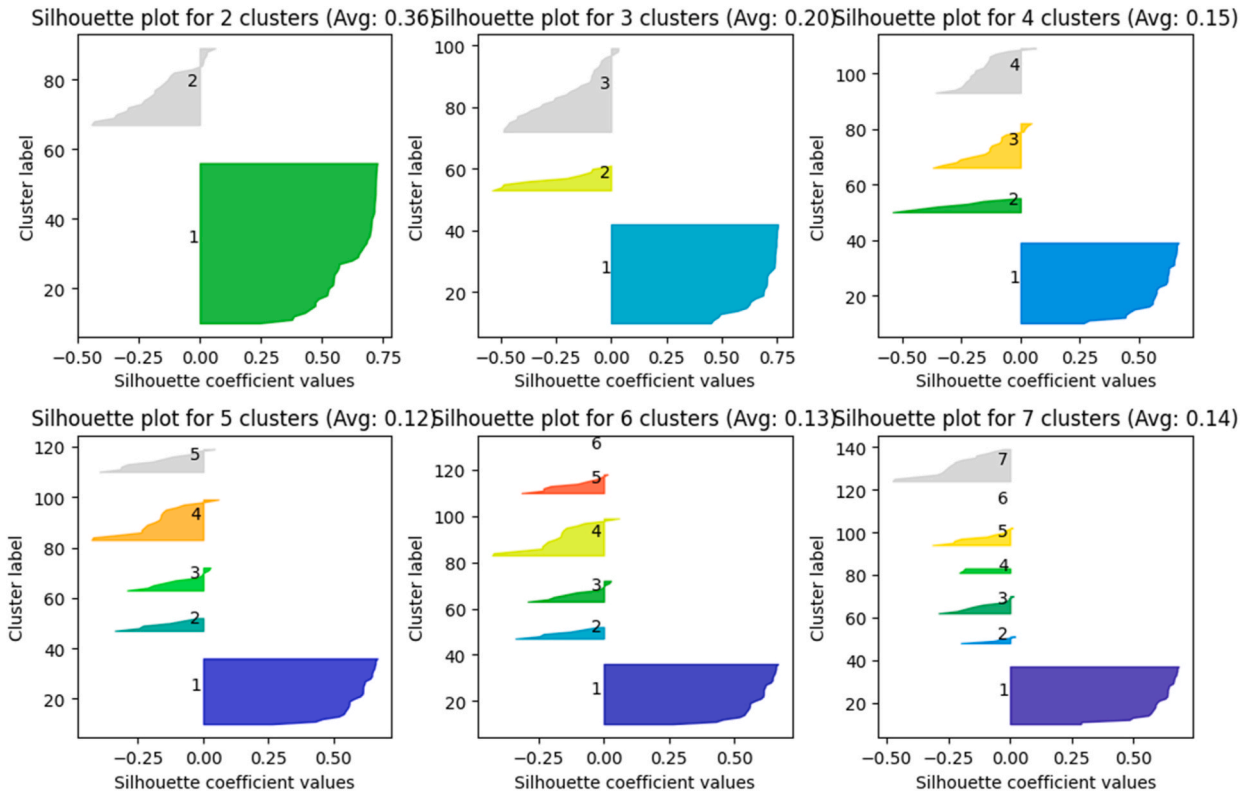

Fig. 3. Silhouette plots for cluster 3.

5. Discussion

This study bridges critical theoretical and methodological gaps in CPS research by leveraging machine learning to identify distinct, granular profiles of student behaviors in MCPS. While established CPS models (Isaksen et al., 2010; Lee et al., 2023; Mumford et al., 1991; Treffinger et al., 2008, 2023) outline abstract problem-solving stages, empirical tools to dynamically map how learners enact these stages—particularly in real-world problem context—remain underdeveloped. Similarly, data-drive approach seldom integrates CPS theories to interpret behavioral patterns at a fine-grained level. Furthermore, we identified and examined differences in the behavioral patterns of typical clusters from both an ability level and cognitive process perspectives at a fine-grained level.

First, three distinct student profiles were identified by K-means clustering analysis based on performance across seven dimensions of MCPS (Tan, 2024), revealing differences of different groups in abilities demonstrated at various stages of problem finding and solving, as well as in metacognition. Second, time-series K-means clustering revealed two distinct sub-clusters within each of above three clusters based on the time-action sequence dataset, resulting in six sub-clusters with different characteristics in problem-solving process. Lastly, LSA further uncovered the differences in behavioral patterns of these six sub-clusters, such as cognitive efficiency, metacognitive regulation, behavioral adaptability, and motivational profiles. HMM results complemented LSA by revealing the latent stage dynamics (e.g., state persistence, transition patterns) and by showing systematic differences between subclusters in terms of state matching and unmatched behavioral states. Hence, the use of a person-centered data-drive approach helped us obtain a more nuanced picture of MCPS patterns. The theoretical and practical implications for enhancing students' MCPS abilities are discussed.

5.1. *Typical clusters of students' ability level in mathematical creative problem solving*

This study found three distinct MCPS ability-based clusters through multi-stage performance analysis, including *Creative Problem Solvers* (Cluster 1), *Underperformers* (Cluster 2) and *Problem Finders* (Cluster 3). Representing 18.64% of the students, *Creative Problem Solvers*, a minority group, demonstrated the highest level of competence in all key cognitive aspects of MCPS, excelling in problem finding and solution generating and evaluating. These advanced problem solvers also possess well-developed essential elements of MCPS, such as mathematical knowledge and skills, divergent and convergent thinkings, and metacognition.

Second, the dominance of *Underperformers* (Cluster 2, 70.44%)—struggling across all CPS dimensions—reveals a substantial gap within the sampled context. While prior work notes CPS challenges (e.g., Sak & Maker, 2006), this study quantifies their prevalence, suggesting a need for instructional support in similar settings. This finding indicates a potential need to reorient instruction toward adaptive problem-solving in the studied environment.

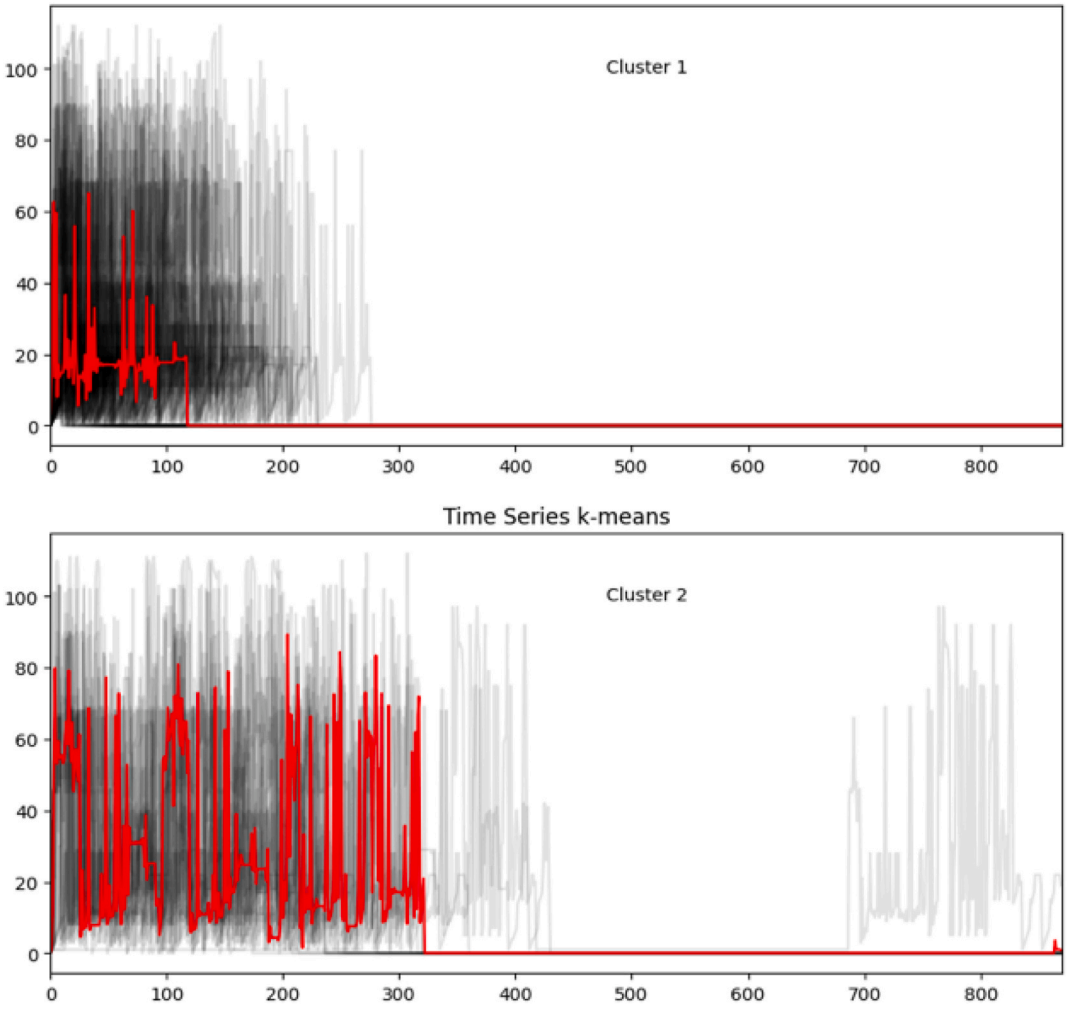

Fig. 4. Time series K-means clustering plots for cluster 1.

Lastly, *Problem Finders* (10.92%) challenges the assumption that strong problem discovery inherently predicts solution competence. This group's profile—excellent problem identification but poor solution generation—reveals a critical decoupling of CPS stages, a phenomenon underexplored in existing literature. This aligns partially with Mumford et al.'s (1991) observation that problem construction does not guarantee effective solution implementation but quantifies it in an educational context.

While foundational CPS models (e.g., Isaksen et al., 2010; Mumford et al., 1991; Treffinger et al., 2008, 2023) conceptualize CPS as sequential stages (e.g., problem finding, ideation, evaluation), prior research seldom empirically segments learners based on differential proficiency across these stages. Our findings extend past theoretical research and variable-centered empirical studies by adopting a person-centered approach to identify different subgroups of students regarding their performance at key stages of MCPS process. Crucially, this finding demonstrates that CPS is not a unitary trait but a multidimensional construct where strengths in one stage do not guarantee proficiency in others, which revealed profound unevenness in individual MCPS development. This finding also highlights that structural comprehension and representation of ill-defined problems is a foundational prerequisite for effective solution generation—yet alone insufficient for successful resolution, which also aligns with theoretical models positing the importance of problem finding in CPS (Abdulla et al., 2020; Mumford et al., 1996; Rubenstein et al., 2020).

It is important to position MCPS not as a reflection of students' existing competencies under traditional instruction, but as an aspirational competency aligned with ongoing curriculum reform (Tang, 2025). The high proportion of low-performing students observed in this study may not solely indicate individual deficits; rather, it may reflect a transitional stage in pedagogical development, where classroom practices have not yet fully adapted to foster creative problem-solving (Johnson & Fitzmaurice, 2025). Accordingly, this study treats observed performance gaps as partly indicative of the gap between curricular aspirations and current instructional realities, rather than attributing them exclusively to student deficiencies. This perspective situates our findings within broader reform efforts and avoids over-pathologizing learners.

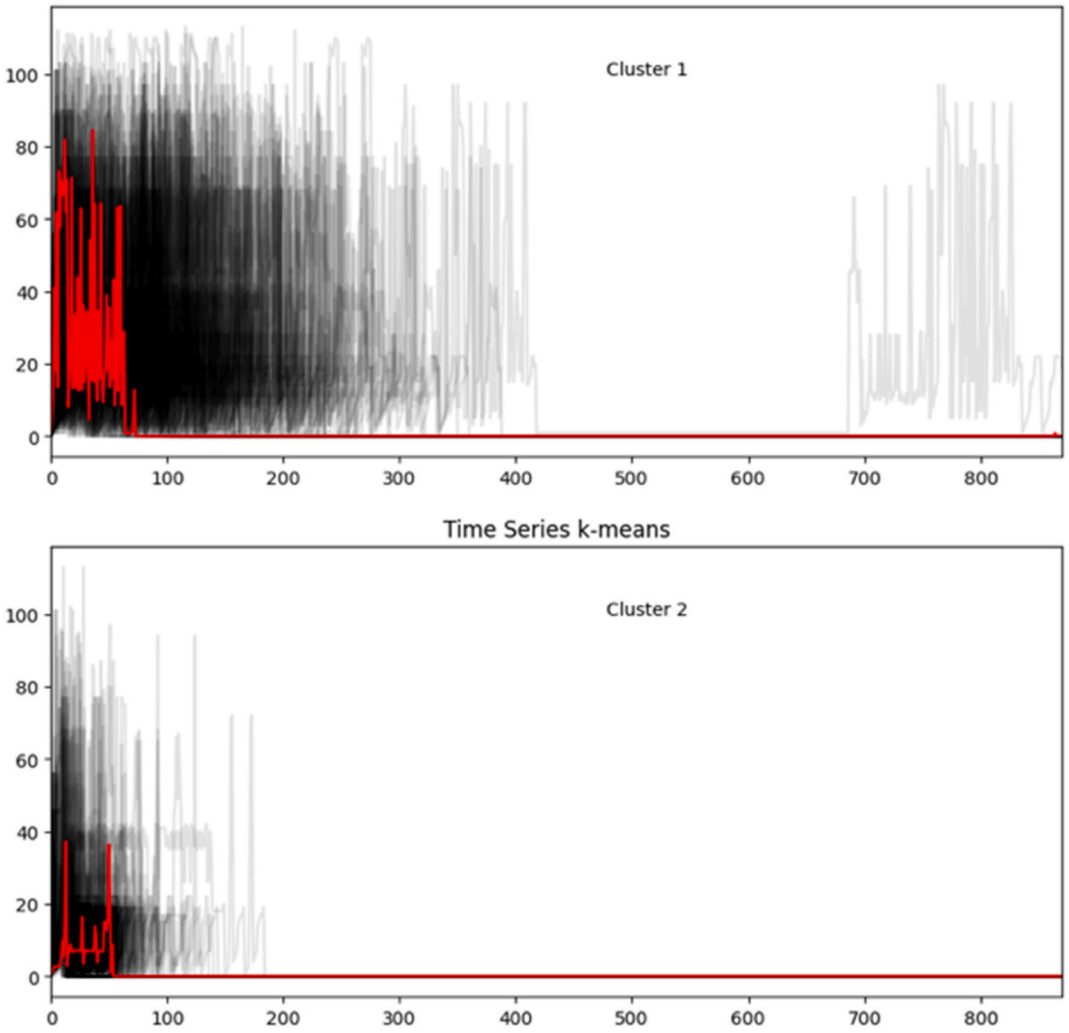

Fig. 5. Time series K-means clustering plots for cluster 2.

5.2. *Typical clusters of students' behavioral pattens in mathematical creative problem solving*

This study further identified two distinct sub-clusters within each ability-based cluster, revealing that students with similar ability levels can follow fundamentally different cognitive-metacognitive processes. HMMs and LSA confirmed these differences: HMMs showed that sub-clusters varied in their optimal number of latent states (5–14) and uncovered unique behavioral states with no counterpart in the paired sub-cluster, while LSA revealed differences in cognitive efficiency, metacognitive regulation, and behavioral adaptability. Together, these findings provide direct evidence that monolithic interpretations of CPS competence are insufficient.

For *Creative Problem Solvers* (Cluster 1), Clusters 1.1 and 1.2 differed in their problem-solving processes and styles. Specifically, Clusters 1.1 (9.45%) was characterized by rapid task completion with short behavioral sequences, minimal backtracking, and high efficiency in identifying critical information and variables, which indicated their high confidence, goal-oriented strategies, and fast decision-making. Consistently, HMM revealed that Cluster 1.1 had a highly stable task-boundary state (state 9, stability = 0.96, duration = 26.9 steps) and its most occupied state (state 9, steady state proportion = 32.8 %) was dominated by task control actions, reflecting efficient stage transitions. However, there is a risk of over-reliance on their existing skills and a potential lack of depth in exploring alternative solutions.

To potentially address this, educators might consider implement strategies that challenge learners with complex, open-ended problems, incorporating structured phases for metacognitive reflection to promote awareness of alternative approaches (Liljedahl et al., 2016; Suriyon et al., 2013). For example, presenting problems with multiple correct solutions or those that involve interdisciplinary knowledge could potentially encourage them to expand their thinking beyond their current comfort zones and counteract overreliance on schema-driven strategies (Liljedahl et al., 2016; Zhang et al., 2024). Additionally, providing opportunities for peer collaboration with Cluster 1.2 might expose them to different problem-solving styles and enhance their ability to consider alternative perspectives (Fawcett & Garton, 2011).

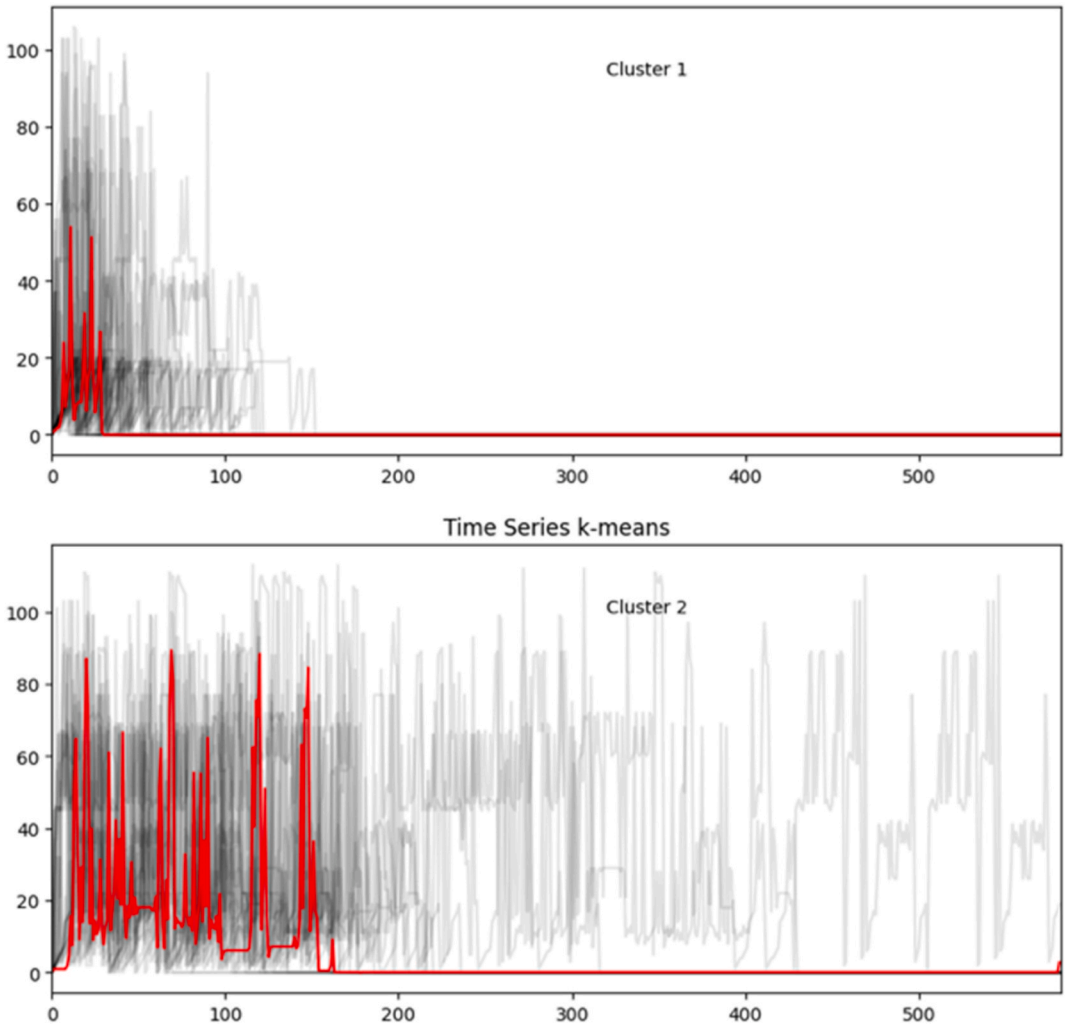

Fig. 6. Time series K-means clustering plots for cluster 3.

Table 8
The summary of behavioral patterns for all clusters.

	Cluster 1.1	Cluster 1.2	Cluster 2.1	Cluster 2.2	Cluster 3.1	Cluster 3.2
Sample size	71 (9.45%)	64 (8.52%)	433 (57.66%)	107 (14.25%)	27(3.60%)	49 (6.52%)
Number of nodes	58	56	58	58	48	52
Number of edges	156	153	189	180	99	125
Weight range of edges	[2, 26]	[2, 30]	[2, 47]	[2, 39]	[2, 23]	[2, 25]
Average degree of edges	5.38	5.46	6.52	6.21	4.13	4.81
Maximum node output	7(*shift21*)	6 (*shift12/22/23/34/44*)	9 (*shift56*)	8 (*shift34*)	5 (*shift23*)	6 (*shift44*)
Average output of getting information (info)	2.56	1.78	4.56	1.89	2.11	2.44
Average output of shifting items (shift)	3.11	3.50	4.00	3.89	2.40	2.70
Average output of submitting answers (submit)	2.17	1.33	1.33	1.50	1.67	1.33
Maximum node input	8 (*shift21*)	7 (*shift12/34*)	8 (*shift11/65*)	7 (*shift44*)	5 (*shift11/12/23*)	7 (*shift12*)
Average input of getting information (info)	2.33	2.22	3.78	2.11	1.89	2.33
Average input of shifting items(shift)	3.11	3.29	3.69	3.69	2.46	2.72
Average input of submitting answers (submit)	2.00	3.00	2.50	3.33	1.83	2.17

Note. The notation "*shift12*" represents the action of transitioning the system's answer interface from assessment item 1 to item 2. Similar notations (e.g., *shift23*, *shift34*) follow the same convention, indicating a navigation from the first numbered item to the second.

Table 9
The summary of behavioral patterns for all clusters.

	Cluster 1.1	Cluster 1.2	Cluster 2.1	Cluster 2.2	Cluster 3.1	Cluster 3.2
Similar behavioral patterns	*submit1→shift12, submit2→shift23, submit3→shift34, submit4→shift45*		*submit1→shift12→submit2→shift23→submit3→shift34, submit4→shift45, submit5→shift56, submit6→taskend, chat→info*		*submit1→shift12→submit2→shift23→submit3→shift34, submit4→shift45, submit6→taskend*	
Different behavioral patterns	*shift14→shift41→shift1, shift15→shift51, shift52→shift24, shift26→shift62*	*shift23→submit3, submit5→shift56, submit6→taskend, chat→info, info3→info4, shift33→shift63, shift36→shift62, shift46→shift66, shift51→shift15→shift53, shift65→shift55*	*shift34→submit4, shift45→submit5, info2→info3→info4→info5, info6→info7, addnote→deletenote, shift26→shift66, shift31→shift13, shift53→shift33, shift54→shift44*	-	*submit5→shift56, shift66→shift61, shift43→shift32*	*chat→info, info6→info7, shift66→shift26, shift61→shift11*

Note. In this table, behavioral patterns are represented as sequences of actions, with arrows (→) indicating the temporal order. The notation follows these conventions.
Shift12: Navigating from item 1 to item 2.
Submit1: Submitting a response for item 1.
Chat: Clicking on the problem scenario in the chat/interaction panel.
Info: Clicking to access information items in the interaction panel.
Addnote/deletenote: Adding or deleting a note.
Sequences such as *submit1→shift12* describe a common chain of actions (e.g., submitting item 1 and then proceeding to item 2).

Table 10
The summary of best performance HMMs for all clusters.

	Cluster 1.1	Cluster 1.2	Cluster 2.1	Cluster 2.2	Cluster 3.1	Cluster 3.2
Best-fitting number of hidden states	10	13	14	10	5	6
LogLik	−6598.57	−11918.66	−19759.57	−18457.1	−5587.99	−4929.03
AIC	14275.13	25265.32	41141.14	37992.14	11563.97	10384.06
BIC	17850.71	30680.03	47399.56	42191.67	12813.02	11989.6
Stability range	0.68–0.96	0.27–0.96	0.49–0.96	0.00–0.98	0.71–0.95	0.86–0.96
Mean duration range (steps)	3.07–26.87	1.37–22.29	1.96–23.51	1.00–64.23	3.41–20.46	6.92–24.13
Most persistent state (stability/duration)	S9 (0.96/26.87)	S11 (0.96/22.29)	S8 (0.96/23.51)	S7 (0.98/64.23)	S2 (0.95/20.46)	S4 (0.96/24.13)
Most occupied state (steady-state proportion)	S9 (32.8%)	S10 (18.1%)	S2 (14.8%)	S7 (17.0%)	S2 (34.3%)	S4 (46.4%)
Dominant actions in top states	*symbolic, taskstart/taskend, shift31*	*info, shift46/53/54/55/64, submit3*	*symbolic, shift11/12/14/26/51/52/53*	*info, shift12/23/45, submit2/4/6*	*shift23/51*	*shift61/66*

Note. To reduce model complexity, several raw actions were merged into higher-level categories. The category *symbolic* combines *addnote*, *deletenote*, *addsymbol*, and *calculate*. The category *info* merges *chat* and *info1* through *info8*.

Table 11
State matching and dynamic differences between clusters 1.1 and 1.2.

Matched state of Cluster 2.1	Matched state of Cluster 2.2	Cosine distance	Stability difference (ΔStab)	Mean duration difference (ΔDur, steps)	Steady-state difference (ΔSteady)
S1	S4	**0.16**	+0.104	+5.07	+0.039
S2	S3	0.79	−0.129	−5.91	+0.046
S3	S1	0.87	+0.311	+1.50	−0.004
S4	S12	0.80	+0.131	+2.39	+0.076
S5	S13	1.00	+0.129	+0.89	−0.068
S6	S8	**0.17**	+0.529	+3.64	+0.030
S7	S2	1.00	−0.066	−2.39	−0.024
S8	S10	1.00	−0.198	−16.62	−0.064
S9	S5	1.00	+0.065	+17.07	+0.262
S10	S11	**0.21**	−0.195	−18.13	−0.048

Note. Differences are calculated as Cluster1.1 minus Cluster1.2.
Bold values indicate close matching (small distance).
Unmatched states in Cluster1.2: S6 (*shift45, info, shift64*), S7 (*shift55, shift33*), S9 (*shift53*).

Table 12
State matching and dynamic differences between clusters 2.1 and 2.2.

Matched state of Cluster 2.1	Matched state of Cluster 2.2	Cosine distance	Stability difference (ΔStab)	Mean duration difference (ΔDur, steps)	Steady-state difference (ΔSteady)
S1	S4	1.00	−0.731	−2.71	+0.049
S2	S1	0.49	−0.064	−2.20	+0.006
S3	S13	0.81	−0.938	−15.13	+0.017
S4	S3	**0.12**	+0.095	+3.51	−0.033
S5	S5	1.00	+0.326	+4.87	−0.041
S6	S7	0.67	+0.385	+5.96	+0.100
S7	S2	0.51	+0.176	+59.01	+0.022
S8	S6	1.00	−0.021	−0.19	+0.031
S9	S8	1.00	−0.144	−18.16	−0.014
S10	S9	1.00	+0.443	+12.51	+0.037

Note. Differences are calculated as Cluster2.2 minus Cluster2.1.
Bold values indicate close matching (small distance).
Unmatched states in Cluster2.1: S10 (*shift51*), S11 (*shift52*), S12 (*shift61, shift62, shift52, shift66*), S14 (*shift33, shift16*).

In contrast, Clusters 1.2 (8.52%) took more time to complete tasks, frequently reevaluate information, and refine their solutions iteratively, which indicated their methodical patience, deliberate analysis, and high level of metacognitive regulation. Behaviorally, they exhibit systematic exploration, balancing analytical depth with adaptive recalibration—traits aligned with expertise development but potentially inefficient in time-constrained settings. This subcluster required 13 latent states (the highest among all groups), and its most occupied state (state 10, steady state proportion = 18.1 %) was dominated by *shift*64, a specific backtracking action. The state-matching analysis showed that Cluster 1.2 possessed one state (S6) unmatched to any state in Cluster 1.1, characterized by mixed

Table 13
State matching and dynamic differences between clusters 3.1 and 3.2.

Matched state of Cluster 3.1	Matched state of Cluster 3.2	Cosine distance	Stability difference (ΔStab)	Mean duration difference (ΔDur, steps)	Steady-state difference (ΔSteady)
S1	S1	0.90	+0.022	+0.93	−0.023
S2	S4	0.52	−0.010	−0.51	−0.020
S3	S6	1.00	+0.062	+3.17	−0.051
S4	S5	0.89	**+0.252**	**+20.72**	**+0.437**
S5	S2	1.00	−0.084	−12.92	−0.210

Note. Differences are calculated as Cluster3.1 minus Cluster3.2.
Positive values indicate higher values in Cluster3.1. Bold numbers highlight the largest absolute differences. Unmatched state in Cluster3.2: S3 (*submit5, info, shift52, shift54*).

shift45, info, and shift64 actions – reflecting iterative re-evaluation not present in the rapid solvers. Educational support could aim to strike a balance between maintaining their depth of analysis and improving their efficiency. This could involve training in time management techniques specifically tailored to problem-solving tasks, as well as exposure to more time-constrained practice scenarios (Najdowski, 2017; Suriyon et al., 2013). By doing so, they could potentially develop the ability to make more informed decisions within a reasonable time frame without sacrificing the quality of their problem-solving. Moreover, pairing with Cluster 1.1 peers to merge rapid execution with iterative refinement also could foster their symbiotic learning (Fawcett & Garton, 2011).

For *Underperformers* (Cluster 2), two sub-clusters 2.1 and 2.2, displayed systemic deficits across all key stages. Clusters 2.1 (57.66%) engaged with tasks for a prolonged period but often exhibited disorganized backtracking and a poor linkage between information and tasks. Despite high motivation, their divergent-convergent thinking imbalance led to unproductive persistence, as seen in frequent but aimless behavioral shifts. This aligned with cognitive overload theories, where excessive exploration without strategic filtering hindered progress (Bruno et al., 2024). HMM showed that Cluster 2.1 had the largest number of latent states (14), indicating high behavioral fragmentation. Its most occupied state (state 2, steady proportion = 14.8 %) combined *shift61 and submit6*, reflecting inefficient switching between shift and submission actions. Four states in Cluster 2.1 (S10, S11, S12, S14) had no counterpart in Cluster 2.2, suggesting unique but unproductive latent patterns such as isolated shift actions without strategic integration.

To improve their performance, educational support could focus on visual organizers to systematically decompose problems into manageable components, reducing cognitive overload and enhancing strategic focus (Bruno et al., 2024). By providing explicit instruction and guided practice in these areas, they may learn to approach problems in a more systematic and organized manner. Additionally, a key educational action for this group could be focused on strengthening their divergent and convergent thinking skills. The constrained brainstorming technique was an effective way to address divergent-convergent thinking imbalances, such as requiring learners to pre-identify three viable strategies before problem-solving, which could filter unproductive paths while preserving exploratory flexibility (Cropley et al., 2001; Liljedahl et al., 2016). Lastly, fostering a growth mindset and providing emotional support may help them overcome the frustration associated with their initial struggles and build confidence in their abilities (He & Chiang, 2024).

Clusters 2.2 (14.25%), on the other hand, exhibited metacognitive apathy, marked by cursory engagement and minimal information processing. What is more, their short task durations and low backtracking frequency signal superficial heuristic reliance, likely driven by low self-efficacy or task-value perception. Consistent with this, HMM revealed that Cluster 2.2 had a state (S7) with the highest stability (0.98) and longest mean duration (64.2 steps) across all subclusters, dominated by submit6 – indicating a highly persistent but shallow "submitting" state that traps students without further exploration. The state-matching analysis showed that Cluster 2.2 lacked counterpart states for several of Cluster 2.1's latent patterns, confirming its disengaged nature. Educational interventions for this group could start by identifying the underlying factors contributing to their disengagement, such as a lack of interest in the subject matter or a perceived lack of relevance (Gumaling, 2023). To address this, designing low-stakes, game-based tasks (e.g., puzzle sequences with incremental rewards), and incorporating real-world examples and problem-solving tasks that are relevant to their lives may increase their motivation and self-efficacy, fostering deeper cognitive investment (Bruno et al., 2024). Additionally, deconstructing problems into foundational sub-tasks with immediate feedback and rewards for their efforts (e.g., schema-based instruction for arithmetic operations), may help build procedural fluency, reinforce positive behaviors, and gradually increase their persistence and metacognitive awareness (Suriyon et al., 2013).

For *Problem Finders* (Cluster 3), Clusters 3.1 and 3.2 revealed a dissociation between problem finding and solving. Specifically, Clusters 3.1 (3.60%) completed tasks relatively quickly but had slow behavioral shifts, engaged in passive information processing, and exhibited limited flexibility, which showed their cognitive rigidity, low task motivation, and formulaic solution approaches. HMM indicated that Cluster 3.1 had the fewest latent states (5) among all subclusters, and its most occupied state (state 2, steady proportion = 34.3 %) was dominated by *shift51*, a single shift action – reflecting rigid, repetitive behavioral loops. The state-matching analysis showed that one state in Cluster 3.2 (S3) was unmatched, characterized by *submit5, info, and shift52/54*, representing more varied but still inefficient exploration that Cluster 3.1 lacked.

To help them overcome these limitations, educational interventions could focus on promoting creativity and cognitive flexibility (Cropley et al., 2001; Kwon et al., 2006). This could involve exposing them to a variety of problem-solving strategies and encouraging them to think outside the box. After that, providing opportunities for them to work on projects that require them to apply their knowledge in new and innovative ways may help increase their motivation and engagement (Suriyon et al., 2013). By challenging their existing thought patterns and encouraging them to explore alternative solutions, they could develop the ability to adapt to different

problem-solving situations (Kwon et al., 2006).

Clusters 3.2 (6.52%), in contrast, exhibited persistent but inefficient exploratory behaviors, characterized by frequent backtracking and solution iteration, where high effort intensity coupled with poor strategic calibration resulted in suboptimal problem-solving outcomes. Obviously, their trial-and-error approach, while effortful, lacks metacognitive calibration to prioritize effective strategies. HMM revealed that Cluster 3.2's most occupied state (state 4, steady proportion = 46.4 %) was dominated by *shift24*, a single shift action, but also had several unmatched states (S6, S7, S9) with diverse actions such as *shift45*+*info*+*shift64* and *shift55*+*shift33*, indicating that even though the most frequent state was narrow, the subcluster retained latent capacity for broader exploration that was not strategically deployed.

Educational interventions could focus on teaching them how to assess the feasibility and effectiveness of different strategies before implementing them (Cropley et al., 2001). This could involve providing explicit instruction on problem-solving heuristics and decision-making models, as well as opportunities for them to practice applying these strategies in a structured environment (Luchini et al., 2023). By helping them develop a more strategic approach to problem-solving, they may make more efficient use of their time and effort and achieve better outcomes.

Collectively, this study extends CPS theory by revealing that ability-level homogeneity masks profound process heterogeneity and addresses a critical need in CPS research: to examine cognitive processes at finer temporal and behavioral resolution. While traditional CPS assessments rely on product quality or self-reports (Acar & Runco, 2023; Baer, 2011), computer-based approach—capturing micro-level action sequences—combined with data mining techniques (e.g., time-series clustering analysis, LSA, and HMMs) provides unprecedented empirical evidence of how students navigate CPS stages. This methodology moves beyond static ability metrics to capture dynamic decision-making under uncertainty—the core of real-world CPS.

5.3. Theoretical and practical implications

This study is theoretically important. First it provides empirical evidence that MCPS manifests as heterogeneous behavioral configurations rather than a linear developmental continuum. By identifying distinct student profiles based on performance across multiple stages (problem finding, solution generation, evaluation), the research demonstrates that CPS is not a unitary trait but a multidimensional construct where strengths in one stage do not guarantee proficiency in others. Crucially, the finding that over 70% of students were clustered as "Underperformers" (Cluster 2) offers evidence within the sampled context that difficulties are not isolated but pervasive, affecting coherent progression across multiple CPS stages for a majority of students. This challenges linear, stage-sequential assumptions and provides an empirical anchor for theoretical models positing the complex, non-linear nature of CPS.

Second, the findings reveal that ability level and process pattern are related but not identical dimensions of MCPS. The two-stage clustering approach shows that students with comparable overall ability can exhibit fundamentally different cognitive-metacognitive processes (e.g., efficient vs. reflective; persistent vs. disengaged). This distinction between what students can achieve (ability) and how they achieve it (process) extends existing theories by highlighting that ability-level homogeneity masks profound process heterogeneity—a theoretical insight that would be obscured by variable-centered analyses alone.

Third, the study identifies distinct strategic archetypes in mathematical creative problem solving. The six sub-clusters represent qualitatively different problem-solving styles: for example, rapid goal-oriented strategies (Cluster 1.1), deliberate iterative refinement (Cluster 1.2), unproductive persistence (Cluster 2.1), superficial disengagement (Cluster 2.2), formulaic rigidity (Cluster 3.1), and inefficient trial-and-error (Cluster 3.2). These archetypes provide a typology that can inform future theoretical work on how strategic heterogeneity emerges within the same ability level.

Beyond these theoretical implications, the study also demonstrates a methodological contribution: computer-based tasks combined with machine learning techniques (K-means clustering and LSA) can effectively capture time-dependent behavioral data, enabling dynamic analysis of CPS processes. This opens new avenues for understanding how students enact problem-solving strategies and for developing adaptive instructional approaches. Future studies may build on these theoretical and methodological foundations to investigate stage interactions, test targeted interventions, and incorporate multimodal data (e.g., eye-tracking, verbal protocols).

This study has several practical implications. The quantitative predominance of the "*Underperformer*" profile (Cluster 2) suggests potential implications for educational practice within the sampled context. It signals a notable gap in students' preparedness for creative problem-solving in this sample and offers preliminary empirical evidence that may inform instructional adjustments aimed at fostering higher-order thinking in similar educational settings.

First, this study reveals that many students (e.g., Cluster 1.1's limited solution exploration, Clusters 2.1's unproductive persistence, and Clusters 3.2's inefficient trial-and-error), regardless of their ability level, struggle with monitoring and adjusting their problem-solving strategies. This finding suggests that incorporating explicit metacognitive activities into instruction—such as structured reflection checkpoints (e.g., reflective journaling, "stop-and-check" prompts) and decision-making templates—may be beneficial (Najdowski, 2017; Suriyon et al., 2013). If effectively implemented, such practices could potentially cultivate students' self-regulation, leading to more effective and flexible problem-solving approaches in real-world contexts (Zimmerman, 2002). However, the effectiveness of these strategies requires future experimental validation.

Second, Clusters 3.1's formulaic solution attempts and Clusters 2.1's difficulty abandoning ineffective strategies reveal overreliance on familiar approaches. The observed overreliance on familiar methods and difficulty adapting to new problem types highlight the potential value of instructional strategies that foster cognitive flexibility (Krems, 2014). Designing tasks that require students to explore multiple solutions, challenge standard approaches, and transfer knowledge across domains can help students develop cognitive flexibility vital for CPS beyond the classroom (Lee et al., 2024).

Third, Cluster 2.2's disengagement and Cluster 3.2's misdirected effort intensity indicates poor calibration between task investment

and strategic value. These findings point to a disconnect between students' motivation and their cognitive efforts. Practical interventions such as gamification, time-bound challenges, and visual progress tracking could potentially motivate students and help them calibrate their effort levels to task demands (Gumaling, 2023). Such approaches might enhance enhances persistence, engagement, and the quality of their problem-solving efforts.

Finally, *Problem Finders* (Cluster 1)'s strong problem discovery but weak solution implementation highlights disconnected skill development. This fragmentation of skills across different problem-solving stages suggests a potential need to integrate learning experiences. Instructional design could consider including stage-bridging activities, peer collaboration, and sequenced tasks that explicitly connect problem discovery, ideation, and evaluation (Ge & Land, 2003). These approaches may promote the development of cohesive problem-solving competence, enabling students to transfer skills more effectively to unfamiliar and complex problems.

Overall, educational interventions may benefit from focusing on developing students' metacognitive skills, fostering cognitive flexibility, aligning motivation with effort, and integrating problem-solving stages through collaborative and sequenced learning activities. It should be noted that these implications are drawn from a cross-sectional, descriptive study; future research is needed to test the effectiveness of such interventions experimentally.

5.4. Limitations and future directions

This study also had several limitations. First, it focused on CPS in mathematics, which may limit the applicability of the findings to other domains or tasks. Future research should explore whether clustering approaches can be generalized across different subjects, such as science, language arts, or social studies, and varied task formats.

Second, the sample in this study lacked diversity in terms of demographic and educational backgrounds, which may affect the generalizability of the results. Subsequent studies should include students from various regions, school types, socio-economic statuses, and cultural contexts to ensure broader applicability.

Third, the clustering approach relied on predefined metrics such as ability and process, which may not capture all relevant behavioral patterns. Advancements in analytical methods, such as machine learning and deep learning, could further enhance the analysis of large-scale, multimodal data to provide richer insights into students' cognitive and non-cognitive abilities.

Fourth, this study used only process data such as clickstream and response patterns; future research should incorporate multimodal data sources—including text, audio, video, and physiological signals—to gain a comprehensive understanding of students' behaviors, emotions, and engagement levels. Such integrative approaches can facilitate the development of more personalized and adaptive learning interventions.

Fifth, this analysis is based on a single assessment point, providing a cross-sectional snapshot of student problem-solving profiles. Consequently, while these profiles offer a diagnostic foundation for targeted support, this study's design does not permit the direct observation of skill development or growth over time. The call for longitudinal research is therefore essential to validate the potential of these insights. Future studies are needed to track how students' problem-solving behaviors evolve and to experimentally examine whether interventions informed by such profile-based diagnoses effectively enhance long-term learning outcomes.

6. Conclusion

This study systematically identified and analyzed distinct student behavioral profiles in MCPS within a junior school context by integrating ability-based and cognitive process perspectives through advanced EDM techniques. Three primary clusters emerged: *Creative Problem Solvers* (high competence), *Underperformers* (widespread difficulties, the largest group), and *Problem Finders* (strong problem discovery but weak solution implementation). Each cluster exhibited unique strengths and challenges in problem-solving efficiency, divergent-convergent thinking, metacognitive regulation, and motivational engagement.

The contributions of this research are threefold. First, it provides empirical support for conceptualizing MCPS as a multi-stage process involving interconnected cognitive skills, demonstrating that behavioral profiles vary systematically across stages and ability levels. Second, by leveraging computer-based tasks and EDM techniques (K-means clustering and LSA), the study illustrates how rich, time-dependent process data can be used to dynamically analyze CPS processes, opening new avenues for developing adaptive instructional strategies. Third, the findings reveal significant individual differences in problem-solving behaviors even among students with similar ability levels, underscoring the necessity for personalized learning environments and targeted interventions that foster metacognitive skills, cognitive flexibility, and calibrated effort. Collectively, these insights contribute to both the methodological advancement of CPS assessment and offer a potential analytical basis for differentiated instruction, though the effectiveness of such instructional designs requires future empirical testing.

CRediT authorship contribution statement

Yuting Tan: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tao Yang:** Writing – review & editing, Supervision, Funding acquisition.

Ethics approval and consent to participate

The study was reviewed and approved by the research committee in Beijing Normal University, as well as by the committee in local government.

Consent for publication

All the authors have approved the contents of this paper in the Learning and Individual Differences.

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Declaration of competing interest

The authors declare no conflicts of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compedu.2026.105657.

Data availability

Data will be made available on request.

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